

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Evaluates $E(X^2)$ by calculating $1^2 \times 0.2 + 2^2 \times 0.4 + 4^2 \times 0.35 + 9^2 \times 0.05$	AO1.1a	M1	$E(X^2) = 1^2 \times 0.2 + 2^2 \times 0.4 + 4^2 \times 0.35 + 9^2 \times 0.05$ $= 11.45 \text{ or } \frac{229}{20}$
Evaluates $E(Y^2)$ by integrating $\int y^2 \times f(y) \, dy$ Must see the integral	AO1.1a	M1	$E(Y^2) = \int_0^4 y^2 \times \frac{1}{64} y^3 \, dy$ $= \frac{1}{64} \left[\frac{1}{6} y^6 \right]_0^4 = \left[\frac{1}{384} y^6 \right]_0^4 = \frac{32}{3}$
Finds $E(X^2) = 11.45$ and $E(Y^2) = \frac{32}{3}$	AO1.1b	A1	$E(X^2 + Y^2) = E(X^2) + E(Y^2)$ $= 11.45 + \frac{32}{3}$
Uses $E(X^2 + Y^2) = E(X^2) + E(Y^2)$ to show that $E(X^2 + Y^2) = \frac{1327}{60}$ AG Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$= \frac{1327}{60}$
Total 4 marks			

Q2.

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Marking Instructions	AO	Marks	Typical Solution
Obtains w from $\int_0^{15} w l \, dl = 1$ Allow one error if method correct	AO1.1a	M1	$\int_0^{15} w l \, dl = w \left[\frac{l^2}{2} \right]_0^{15}$ $= \frac{225w}{2} = 1$
Obtains $E(L^2)$ by integrating $\int l^2 \times f(l) \, dl$ FT 'their' value for w	AO1.1a	M1	$w = \frac{2}{225}$ Hence
Obtains $E(S)$ by evaluating	AO1.1a	M1	

$\frac{1}{15}(0+1+2+3+4+5)$ Allow one error if method correct			$E(L^2) = \int_0^{15} l^2 \times \frac{2}{225} l$ $= \frac{2}{225} \left[\frac{l^4}{4} \right]_0^{15} = \frac{225}{2}$ $E(S) =$
Uses $E(T) = E(L^2) + \frac{1}{2}E(S)$ (PI)	AO1.1a	M1	$\frac{1}{15}(0 \times 0 + 1 \times 1 + 2 \times 2 + \dots 5 \times 5)$ $= \frac{55}{15} = \frac{11}{3}$ $E(T) = E(L^2) + \frac{1}{2}E(S)$ $= \frac{225}{2} + \frac{11}{6} = \frac{343}{3} = 114\frac{1}{3}$ AG
Shows that $E(T) = 114\frac{1}{3}$ Mark awarded if they have a completely correct solution, which is clear, easy to follow and contains no slips AG	AO2.1	R1	
Total 5 marks			

Q3.

(a) (i) $L = X + Z$

$E(L) = 68 + 73 = \underline{141}$
 CAO

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$V(L) = 10^2 + 15^2 = \underline{325}$
 CAO

B1

2

(ii) $M = X + Y$

$E(M) = 68 + 25 = \underline{93}$
 CAO

B1

$V(M) = 10^2 + 5^2 + 2 \times 10 \times 5 \times (-0.8)$
 Allow 'no 2'

M1

$= 100 + 25 - 80 = \underline{45}$

CAO

A1

3

- (b) (i) Require: $P(L < 150) =$

$$P\left(Z < \frac{150 - 141}{\sqrt{325}}\right)$$

Standardising 150 using
c's $E(L)$ & c's $V(L)$ from (a)(i)

M1

$$= P(Z < 0.5)$$

(0.49923)

$$= \underline{\underline{0.69 \text{ to } 0.692}}$$

AWFW

(0.69119)

A1

2

- (ii) Require: $P(X + Y > 105) = P(M > 105)$

$$= P\left(Z > \frac{105 - 93}{\sqrt{45}}\right)$$

Standardising 105 using
c's $E(M)$ & c's $V(M)$ from (a)(ii)

M1

$$= P(Z > 1.79) = 1 - P(Z < 1.79)$$

Correct area change (1.78885)

May be implied by a **correct** answer
or by an answer < 0.5

m1

$$= \underline{\underline{0.036 \text{ to } 0.038}}$$

AWFW

(0.03682)

A1

3

[10]

Q4.

- (a) (i) $E(X - Y) = 120 - 90 = 30$

CAO

B1

1

(ii) $\text{Var}(X - Y) = 36 + 13 = 49$
CAO

B1

1

(b) $P(X - Y > 40.5)$

$$= P\left(Z > \frac{40.5 - 30}{7}\right)$$

CAO

M1

$$= P(Z > 1.5)$$

On their μ and σ

A1ft

$$= 1 - \Phi(1.5)$$

$$= 1 - 0.93319$$

m1

$$= 0.06681$$

AWRT 0.0668

A1

4

(c) (i) $E(T) = E(X_1) + E(X_2) + 2E(Y)$
 $= 120 + 120 + 180$
 $= 420$
 CAO

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$$\text{Var}(T) = \text{Var}(X_1) + \text{Var}(X_2) + 4\text{Var}(Y)$$

M1

$$= 36 + 36 + 52$$

$$= 124$$

CAO

A1

3

(ii) $T \sim N(420, 124)$

$$P(T < 400) = P\left(Z < \frac{400 - 420}{\sqrt{124}}\right)$$

M1

$$= P(Z < -1.796)$$

A1ft

$$= 1 - 0.96407$$

M1

$$= 0.03593$$

AWFW 0.0355 to 0.0365

A1

4

[13]

Q5.

- (a) (i) $A_1 \sim N(12, 3)$
 $A_2 \sim N(6, 2)$
 $A_3 \sim N(32, 20)$

$$T_A = A_1 + A_2 + A_3 \sim N(50, 25)$$

B1B1

2

- (ii) $P(T_A < 60) = P(Z < 2.0)$

M1

$$= 0.97725$$

$$\left[\frac{60 - \mu}{\sigma} \right] \text{ for their } \mu, \sigma$$

A1f.t.

2

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- (b) $T_B \sim N(53, 16)$
 $P(T_B < 60) = P(Z < 1.75)$
 $= 0.95994$

B1

1

- (c) P(at least one will take > 1 hour)
 $= 1 - 0.97725 \times 0.95994$

M1A1f.t.

$$= 1 - 0.9381$$

$$= 0.0619$$

$$(0.02275 \times 0.95994) +$$

$$(0.04006 \times 0.97725) +$$

$$(0.04006 \times 0.02275)$$

$$= 0.021839 + 0.03915 + 0.00091$$

$$= 0.0619$$

A1

3

[8]

Q6.

(a) (i) $E(M_1 + M_2 + M_3) = 3 \times 1.7 = 5.1$

B1

1

(ii) $\text{Var}(M_1 + M_2 + M_3) = 3 \times 0.81 = 2.43$

B1

1

(b) (i) $E(R) = \sum p \times R$

$$= 2 \times \frac{1}{8} + 4 \times \frac{9}{16} + 6 \times \frac{5}{16}$$

M1

$$= 4\frac{3}{8}$$

4.375 (AWRT 4.38)

A1

$$\text{Var}(R) = \sum p \times R^2 - E^2(R)$$

$$= 4 \times \frac{1}{8} + 16 \times \frac{9}{16} + 36 \times \frac{5}{16} - \left(4\frac{3}{8}\right)^2$$

M1

$$= 20\frac{3}{4} - 19\frac{9}{64}$$

$$= 1\frac{39}{64}$$

1.609375 (AWRT 1.61)

A1

4

(ii) $E\left(\frac{1}{2}R\right) = \frac{1}{2}E(R) = \frac{1}{2} \times 4\frac{3}{8} = 2\frac{3}{16}$

B1f.t.

$$\text{Var}\left(\frac{1}{2}R\right) = \frac{1}{4} \times \text{Var}(R)$$

$$= \frac{1}{4} \times 1 \frac{39}{64}$$

M1

$$= \frac{103}{256}$$

0.40234 (AWRT 0.402)
(on their (b)(i) Var(R) value)

A1f.t.

3

(c) (i) $T = \frac{1}{2}R + M_1 + M_2 + M_3$

$$E(T) = E\left(\frac{1}{2}R\right) + E(M_1 + M_2 + M_3)$$

$$= 2 \frac{3}{16} + 5.1$$

$$= 7.29$$

(AWRT 7.29)

7.2875 (a)(i) + (b)(ii)

$$(if (b)(ii) \quad E\left(\frac{1}{2}R\right) = \frac{1}{2} (E(R) \text{ used.}))$$

B1f.t.

1

(ii) $Var(T) = Var\left(\frac{1}{2}R\right) + Var(M_1 + M_2 + M_3)$

$$= \frac{103}{256} + 2.43$$

(their a(ii) + b(ii))

M1

$$= 2.832$$

(AWRT 2.83)

A1f.t.

2

[12]

Q7.

(a) Total time to complete event

B1

1

(b) (i) $E(X) = 32 + 40 + 58$
 $= 130$

B1

1

(ii) $\text{Var}(X) = 25 + 64 + 144$

M1

$= 233$

A1

2

(c) $P(X > 146) = P\left(Z > \frac{146 - 130}{\sqrt{233}}\right)$

$z = \frac{146 - \text{their } b(i)}{\sqrt{\text{their } b(ii)}}$

M1✓

$P(Z > 1.048)$
 $= 1 - \Phi(1.05)$
 $= 1 - 0.8531$
 $= 0.1469$

(calculator 0.14727)

A1✓

$= 0.147(3\text{sf})$

AWRT (CAO)
 (0.15)

A1

3

(d) $Y = X_1 + X_2 + X_3 \sim N(390, 699)$

B1

$P(Y > 412) = P\left(Z > \frac{412 - 390}{\sqrt{699}}\right)$
 $\text{for attempting } \left(\frac{412 - \text{their } \mu}{\text{their } \sigma}\right)$

M1✓

$= P(Z > 0.832)$
 $= 1 - \Phi(0.83)$
 $= 1 - 0.79673$
 $= 0.20327$

A1✓

$$= 0.203$$

AWRT (CAO)

A1

4

[11]

Q8.

$$\text{Mean} = 20 + 10 + 75 + 10 = 115$$

cao

B1

$$\text{Variance} = 3^2 + 3^2 + 10^2 + 2^2$$

adding variances

M1

$$= 122$$

cao; ($\sigma = 11.0$ to 11.1 awfw)
(M0 A0 for $\sigma = 18$)

A1

$$P(J < 120) = P\left(Z < \frac{120 - 115}{\sqrt{122}}\right) =$$

standardising 120 using ft (μ & σ)

M1

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$$P(Z < 0.453) = 0.673 \text{ to } 0.677$$

Awrt

A1

[5]

Q9.

$$C \sim N(1085, 18^2) \quad B \sim N(420, 6^2)$$

$$(a) \quad W = B + \Sigma C$$

$$\text{Mean, } \mu_w = 420 + 6 \times 1085 = \mathbf{6930}$$

cao; accept 6.93

B1

$$\text{Variance, } \sigma_w^2 = 6^2 + 6 \times 18^2$$

Adding (7) variances

M1

= 1980

cao; accept 0.00198

(SD = 44.5 or 0.0445 $\Rightarrow$ A0)

A1

3

(b) $P(W < 7000) = P\left(Z < \frac{7000 - 6930}{\sqrt{1980}}\right)$

Standardising (7000 or 7) using consistent units and ft (μ & σ), not σ^2

M1

= $P(Z < 1.57) = \mathbf{0.941 \text{ to } 0.943}$
awfw (0.94179)

A1

2

[5]

Q10.

$X \sim N(220, 20^2) \quad Y \sim N(175, 40^2)$

$T = X + Y$ has:

mean = 395

cao

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B1

and

variance 2000

cao; accept sd = 44.7 awrt

B1

$P(T < 300) = P\left(Z < \frac{300 - 395}{\sqrt{2000}}\right) =$

standardising 300 using their μ and their σ

M1

$P(Z < -2.12) = \Phi(-2.12) = 1 - \Phi(2.12)$

attempted area change

m1

$$= 0.0165 \text{ to } 0.0170$$

awfw; accept 0.017

A1

[5]

Q11.

(a) $C \sim N(26.8, 0.9^2)$

Total, $T = \sum_{cao}^{25} C$ has mean, $\mu = 670$

B1

and variance, $\sigma^2 = 25 \times \text{Var}(C) =$
Use of, or equivalent

M1

$$20.25$$

awfw 20.2 to 20.3 or $\sigma = 4.5$

A1

$$P(T < 675) = P\left(Z < \frac{675 - 670}{4.5}\right) =$$

Standardising using μ and σ

M1

$$P(Z < 1.11) = 0.866 \text{ to } 0.867$$

awfw

A1

5

(b) Difference, $D = C_1 - C_2$ has

Attempt at distribution of D

M1

Mean, $\mu = 0$

and

Variance, $\sigma^2 = 0.9^2 + 0.9^2 = 1.62$
cao, both

A1

$$P(D > 1) = P\left(Z > \frac{1-0}{\sqrt{1.62}}\right) =$$

Standardising using μ and σ

M1

$$P(Z > 0.786) =$$

$$1 - P(Z < 0.786) = 1 - \Phi(0.786) =$$

Area change

m1

$$1 - 0.784 = 0.216$$

$$P(|D| > 1) = 2 \times P(D > 1) =$$

Multiplication by 2

m1

$$0.429 \text{ to } 0.437$$

awfw

A1

6

[11]

Q12.

$$H \sim N(130, 15^2) \text{ and } R \sim N(115, 10^2)$$

$$T = \sum_{i=1}^{15} H_i + \sum_{i=1}^{35} R_i$$

has

$$\text{mean, } \mu = 5975$$

cao

B1

and

$$\text{variance, } \sigma^2 = 6875$$

cao; accept

B2

$$P(T > 6000) =$$

$$P\left(Z > \frac{6000 - 5975}{\sqrt{6875}}\right) =$$

standardising 6000 using their μ and their σ (not σ^2)
not $(\mu - 6000)$

M1

$P(Z > 0.30) = 1 - \Phi(0.30) =$
attempted area change

m1

0.378 to 0.384
awrt; accept 0.38

A1

[6]



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Examiner reports

Q1.

The majority of students scored three marks for this question but lost the final mark for not giving a complete, clear, easy to follow solution. It was common for students to not make it clear that their final line of working was for $E(X^2 + Y^2)$. Some students calculated $E(X^2)$ and/or $E(Y^2)$ without making it clear that was what was being calculated. Other common errors included finding $E(X)$ and $E(Y)$ and squaring them, or squaring $yf(y)$ within the integral for $E(Y^2)$. Some students put y^2 outside the integral for $E(Y^2)$.

Q3.

In part (a), almost all students answered (i) correctly but in (ii) there was some confusion in the calculation of $\text{Var}(M)$ usually through the omission of ' $\times 2$ ' or the careless use of 15 instead of 5. Students correctly linked part (b)(i) to part (a)(i) and so scored the 2 marks available although there were no marks for the use of 149 or 149.5 instead of 150. Fewer students linked part (b)(ii) to part (a)(ii) but, where they did, then they usually combined the information with 105 (2.00 pm to 3.45 pm) to score 3 marks.

Q4.

It was very encouraging to see so many fully correct solutions to this question. Although the usual errors in calculating the variance were evident, these were less prevalent than in previous series.

Q5.

There were many excellent attempts at parts (a) and (b) with full marks often being gained.

Part (c) seemed to cause many candidates problems with only a few fully correct solutions seen. This was very disappointing at this level. Those candidates who drew a tree diagram had most success although this was obviously not necessary in order to answer the question correctly.

Q6.

The answers seen to this question were much improved on those seen to similar questions on previous papers.

In part (a)(i) most candidates gained the correct answer of 5.1. Although there were many correct answers of 2.43 seen in answers to part (a)(ii), there was a considerable number of candidates who thought that the required variance could be found by using $\text{Var}(M_1 + M_2 + M_3) = 3^2 \times \text{Var}(M)$.

Most candidates calculated $E(R)$ and $\text{Var}(R)$ correctly with only a few thinking that

$\text{Var}(R) = E(R^2)$. There were many correct solutions to part (b)(ii), with $E\left(\frac{1}{2}R\right) = \frac{1}{2}E(R)$

and $\text{Var}\left(\frac{1}{2}R\right) = \frac{1}{4}\text{Var}(R)$ often seen. However, some candidates ignored the 'hence' and

used the distribution for $\frac{1}{2}R$ together with methods already employed in part (b)(i) in order to find the answers.

The vast majority of candidates knew what was expected of them in part (c) and many fully correct answers were seen.

Q7.

Part (a) was usually well attempted but there were some strange answers. For example, “total time to complete the swim by all three random variables” indicated a lack of understanding of what is meant by a random variable.

Part (b)(i) was almost always done correctly, and although there were many correct answers of 233 seen to part (b)(ii) there were some who thought that $\text{Var}(X) = 5 + 8 + 12 = 25$ or that $\text{Var}(X) = (5 + 8 + 12)^2 = 6.25$.

There were many excellent attempts at part (c), even by those candidates who failed to calculate the variance correctly in part (b)(ii). A few candidates used the variance rather than the standard deviation when calculating their z-value.

Although there were some excellent solutions to part (d), the majority of candidates assumed that the total time taken, Y , to be $3X$ when they should have been considering $Y = X_1 + X_2 + X_3$. This made no difference to the value of $\mu = 390$, but led candidates to assume that $\sigma^2 = 9 \times 233 = 2097$ rather than the correct value of $\sigma^2 = 3 \times 233 = 699$. In spite of this, there were some correct methods seen to the remainder of this part of the question. However, only the most able candidates found the correct answer of 0.203.

Q8.

Overall, answers to this question showed a welcome improvement over similar questions on previous papers. Many candidates were able to score full marks although a minority added standard deviations.

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Q9.

In part (a), most candidates evaluated correctly the mean of W as 6930 grams but then most of them made the usual mistake of finding the variance of W as $6^2 + 6^2 \times 18^2$ instead of $6^2 + 6 \times 18^2$.

Q10.

Many candidates scored full or almost full marks on this question. Candidates handled sums and differences of random variables much better than in previous series. In fact, where marks were lost, it was often as a result of incorrect standardisation or using the tables incorrectly, both part of the synoptic assessment. Candidates found correct values for the mean and variance of $X + Y$, then considered 300 (5×60). The small minority who decided to work in minutes divided their variance by 60 rather than 3600.

Q11.

In part (a), virtually all candidates, not surprisingly, obtained a mean for 25 bags of 670 kg, but about half calculated the corresponding variance incorrectly as $(25^2 \times 0.9^2)$. In part (b),

it was pleasing to see the high proportion, relative to previous examinations, of candidates who realised that the difference, D , was distributed $N(0, 1.62)$. Many such candidates then calculated $P(D > 1)$ but missed the fact that this probability had to be doubled.

Q12.

Too many candidates set off down the road of a hypothesis test and so scored few marks.

Candidates attempting this question almost always found the correct value for the mean of $T = \sum H + \sum R$. Correct values for $\text{Var}(T)$ were less common due to the use of $15^2\text{Var}(H) + 35^2\text{Var}(R)$ but, overall, this error was less prevalent than on previous papers. Good candidates then realised that 6000, not 100, had to be standardised and most such candidates undertook this correctly. A small minority of candidates attempted to work in

hours but did not realise that whilst $\frac{5975}{60}$ was correct for the mean, $\frac{6875}{60}$ was incorrect for the variance.



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